

# Lebanese University – Faculty of Engineering III

## Sensors & Instrumentation Project

### Individual Contribution Report

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**Project:** Smart Building Security and Fire-Protection System

### Contribution Summary

From the beginning of the project, I actively participated in brainstorming and developing the project concept. Initially, I suggested a different idea, but after discussing several possibilities with my teammates, I proposed the *Smart Building Security and Fire-Protection System*, which became the final project selected by the team.

After finalizing the concept, I spent several days researching the required hardware components and, with the support of AI-assisted analysis, prepared a complete list of suitable parts. I then purchased all the required components from ElectroLab and began assembling the circuit independently at home. During this stage, I also designed the complete Fritzing circuit diagram before our first full team assembly at the university.

During the hardware implementation, I proposed several practical solutions that helped the team successfully assemble and connect the physical circuit. I participated in wiring the components, integrating the modules, and ensuring that the complete system operated correctly.

I also contributed to the software development by proposing the overall system algorithm and workflow, which served as the foundation for implementing the Arduino firmware with AI assistance. In addition, I created the project's GitHub repository, invited all team members, and organized and uploaded the project documentation, code, datasets, and supporting files.

Throughout the project, I participated in all major meetings, implementation sessions, and testing activities. I assisted during the video demonstration by helping with camera operation, presentation management, and coordinating the demonstration sequence. I also took part in collecting the experimental results, recording the CSV sensor data from the Serial Monitor, organizing the dataset, and contributing to the preparation of the final technical report. Finally, I designed and

prepared the PowerPoint presentation used for the project's 10-minute technical presentation.

## **Declaration**

I certify that the activities described above accurately reflect my individual contribution to the Smart Building Security and Fire-Protection System project.